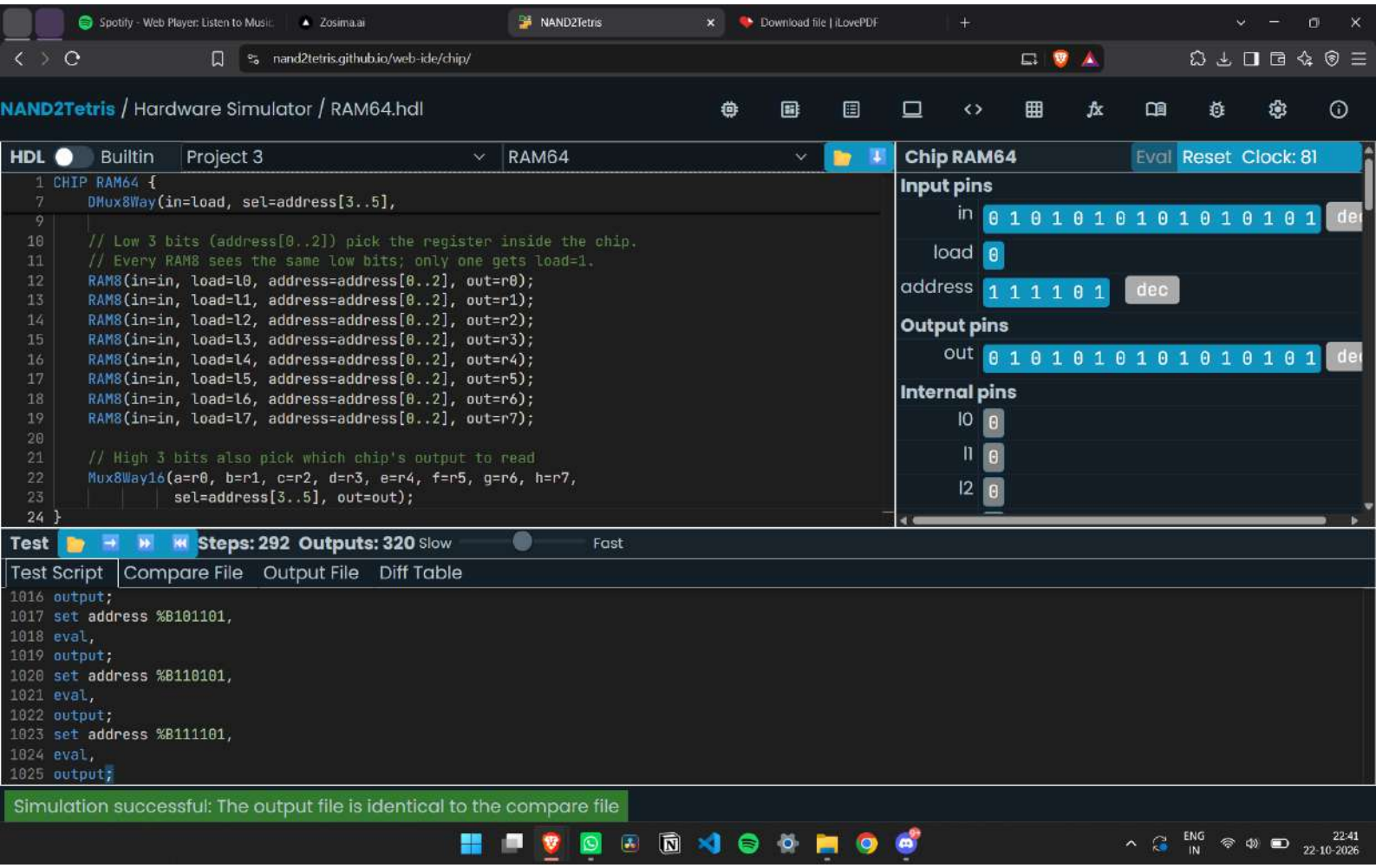
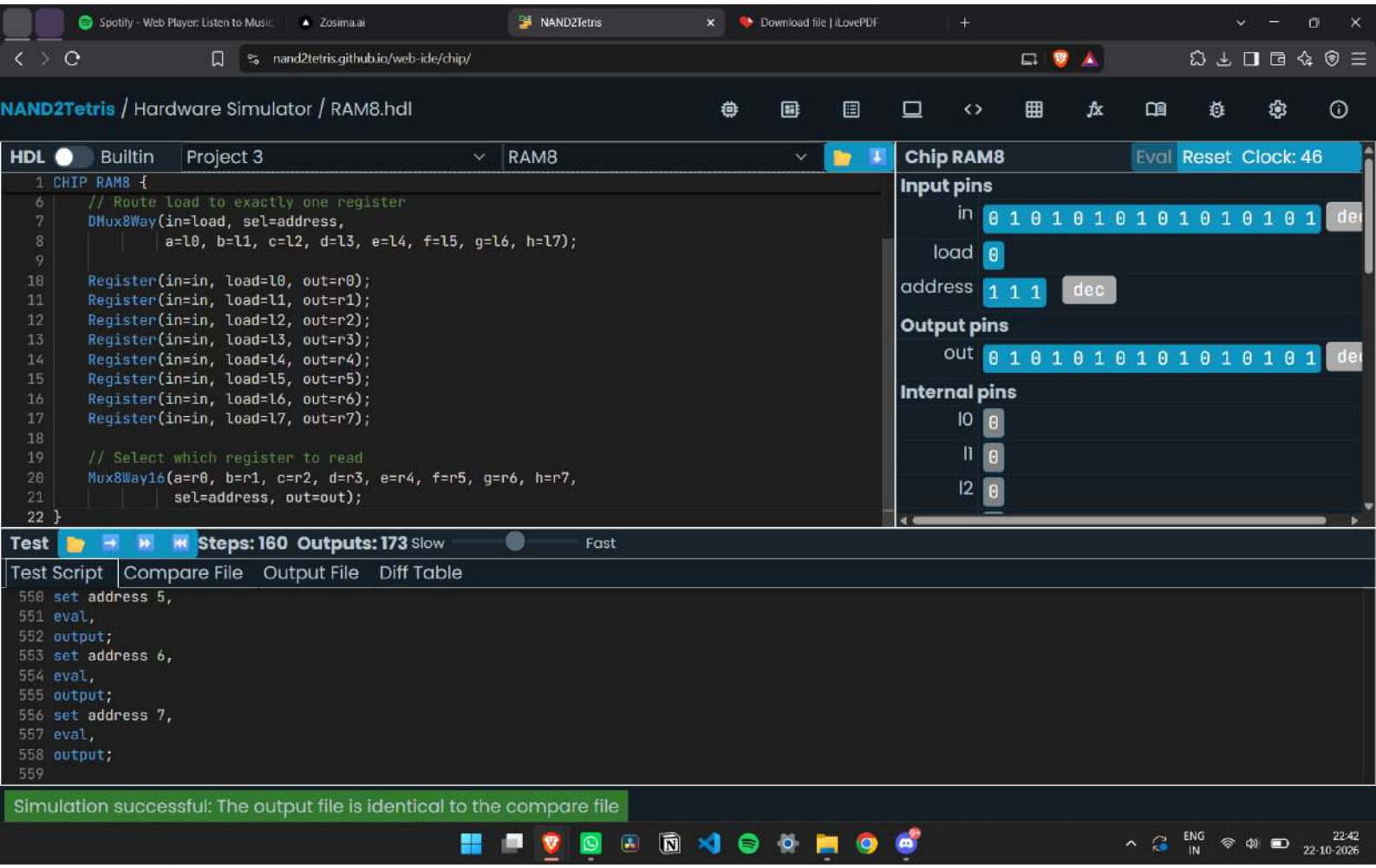
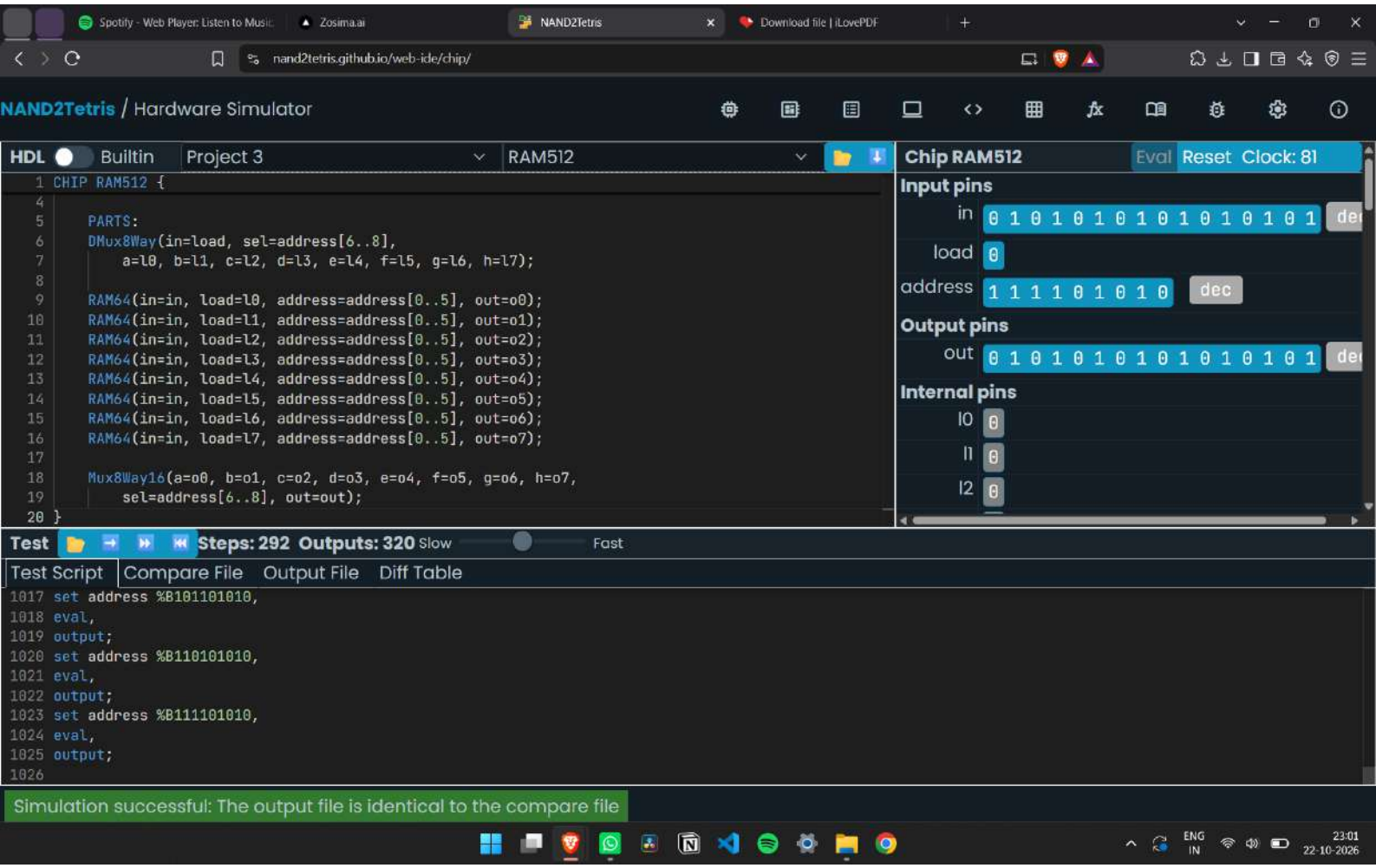
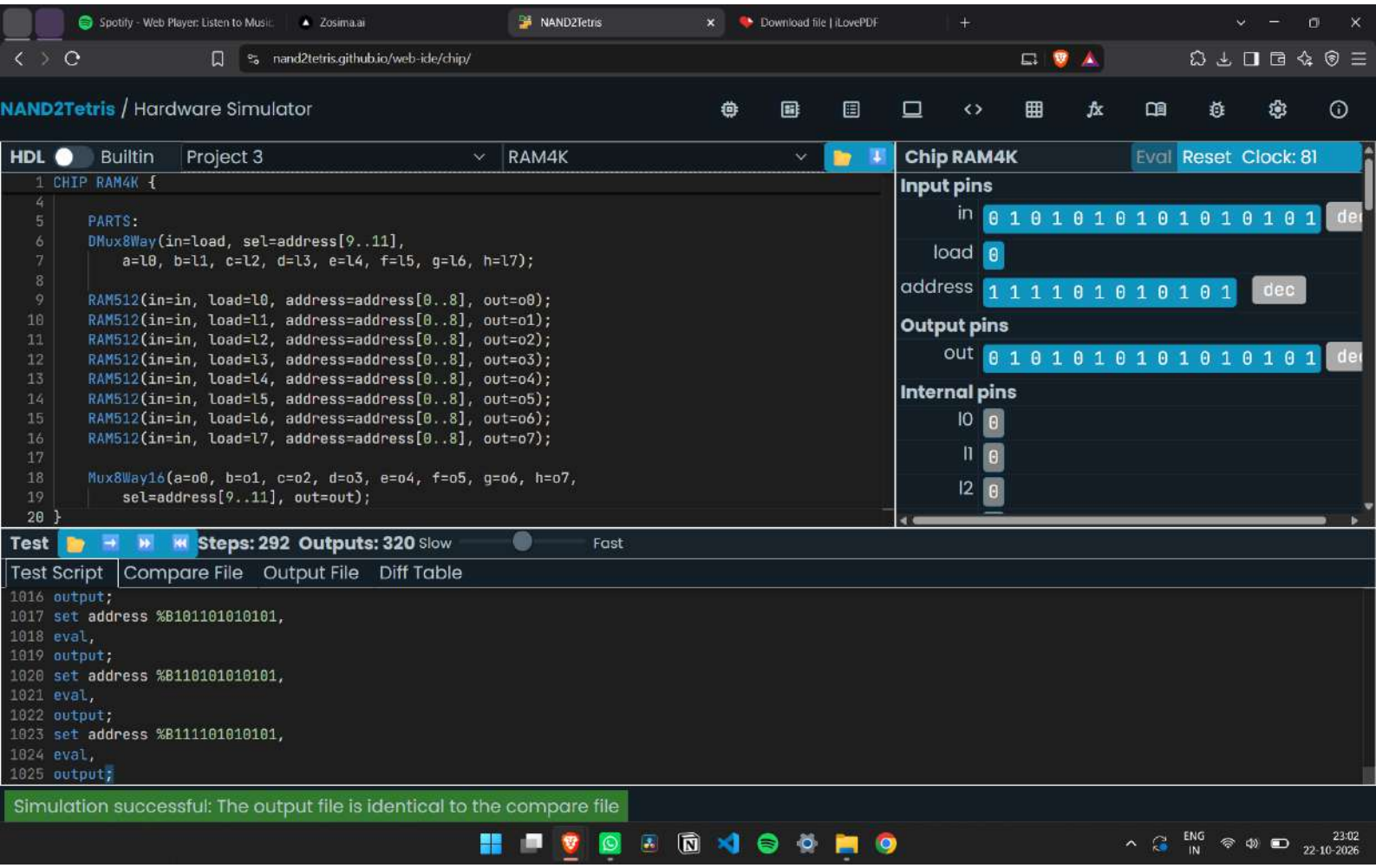
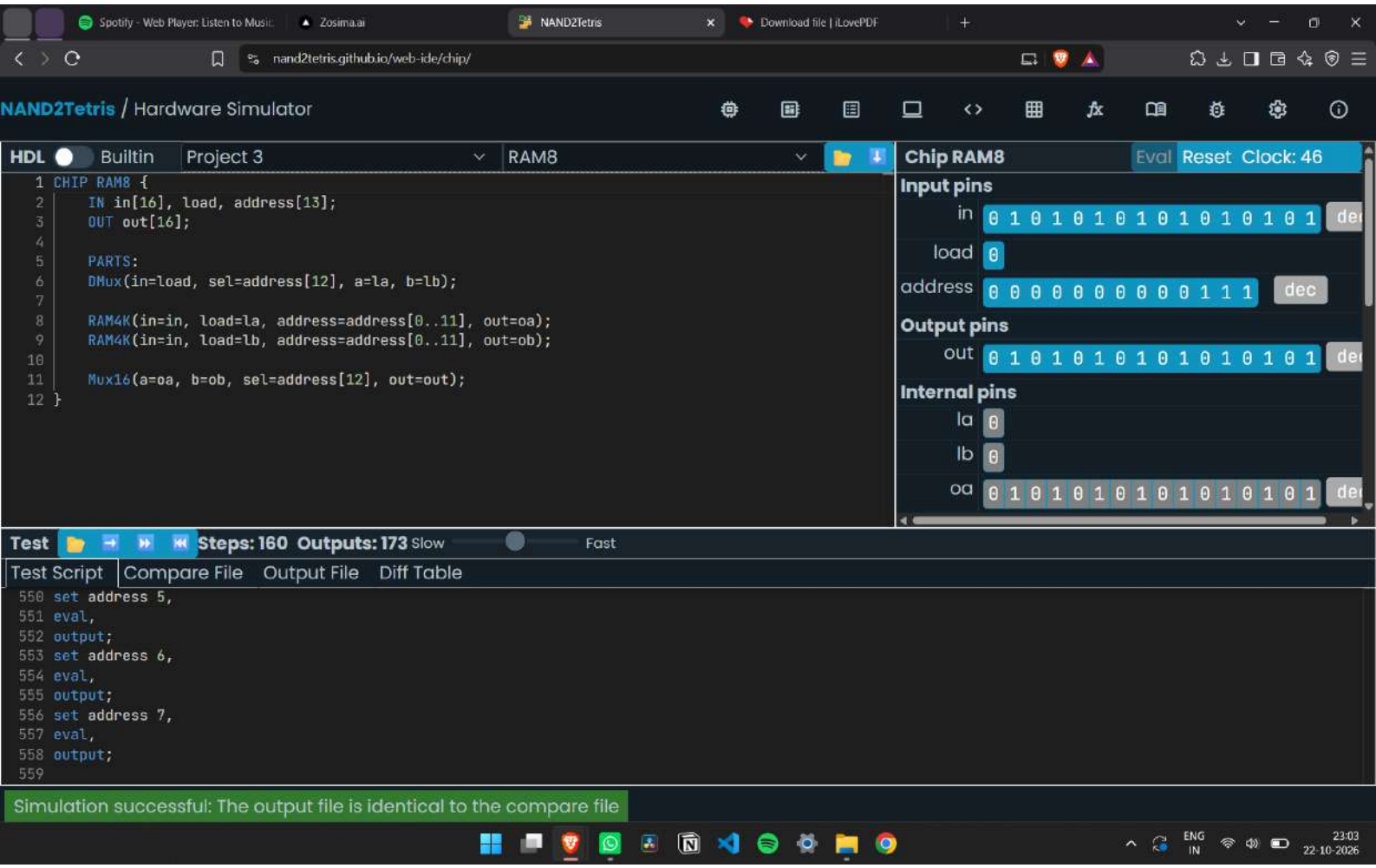
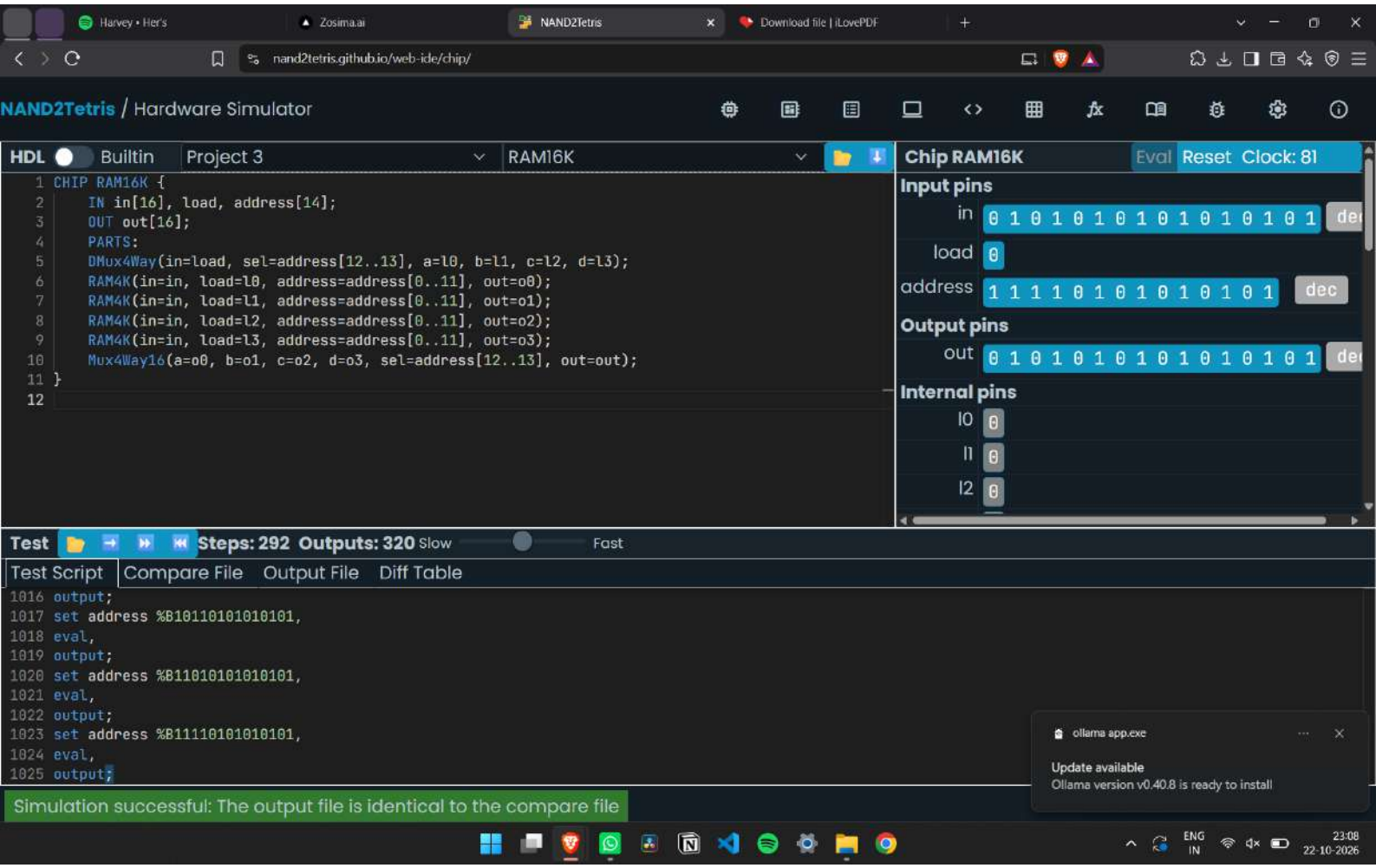

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NAND2Tetris / Hardware Simulator / RAM64.hdl
HDL
Builtin
Project 3
RAM64
1 CHIP RAM64 {
7 DMux8Way(in=load, sel=address[3..5],
9
10 // Low 3 bits (address[0..2]) pick the register inside the chip.
11 // Every RAM8 sees the same low bits; only one gets load=1.
12 RAM8(in=in, load=l0, address=address[0..2], out=r0);
13 RAM8(in=in, load=l1, address=address[0..2], out=r1);
14 RAM8(in=in, load=l2, address=address[0..2], out=r2);
15 RAM8(in=in, load=l3, address=address[0..2], out=r3);
16 RAM8(in=in, load=l4, address=address[0..2], out=r4);
17 RAM8(in=in, load=l5, address=address[0..2], out=r5);
18 RAM8(in=in, load=l6, address=address[0..2], out=r6);
19 RAM8(in=in, load=l7, address=address[0..2], out=r7);
20
21 // High 3 bits also pick which chip's output to read
22 Mux8Way16(a=r0, b=r1, c=r2, d=r3, e=r4, f=r5, g=r6, h=r7,
23 sel=address[3..5], out=out);
24 }
Chip RAM64
Eval
Reset
Clock: 81
Input pins
in
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
load
0
address
1 1 1 1 0 1
dec
Output pins
out
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
Internal pins
l0
0
l1
0
l2
0
Test
Steps: 292 Outputs: 320
Slow
Fast
Test Script
Compare File
Output File
Diff Table
1016 output;
1017 set address %B101101,
1018 eval,
1019 output;
1020 set address %B110101,
1021 eval,
1022 output;
1023 set address %B111101,
1024 eval,
1025 output;
Simulation successful: The output file is identical to the compare file
ENG
IN
22:41
22-10-2026


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HDL
Builtin
Project 3
RAM8
CHIP RAM8 {
// Route load to exactly one register
DMux8Way(in=load, sel=address,
a=l0, b=l1, c=l2, d=l3, e=l4, f=l5, g=l6, h=l7);
Register(in=in, load=l0, out=r0);
Register(in=in, load=l1, out=r1);
Register(in=in, load=l2, out=r2);
Register(in=in, load=l3, out=r3);
Register(in=in, load=l4, out=r4);
Register(in=in, load=l5, out=r5);
Register(in=in, load=l6, out=r6);
Register(in=in, load=l7, out=r7);
// Select which register to read
Mux8Way16(a=r0, b=r1, c=r2, d=r3, e=r4, f=r5, g=r6, h=r7,
sel=address, out=out);
}
Chip RAM8
Eval
Reset
Clock: 46
Input pins
in
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
load
0
address
1 1 1
dec
Output pins
out
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
Internal pins
l0
0
l1
0
l2
0
Test
Steps: 160 Outputs: 173
Slow
Fast
Test Script
Compare File
Output File
Diff Table
set address 5,
eval,
output;
set address 6,
eval,
output;
set address 7,
eval,
output;
Simulation successful: The output file is identical to the compare file
ENG
IN
22:42
22-10-2026


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NAND2Tetris / Hardware Simulator
HDL
Builtin
Project 3
RAM512
CHIP RAM512 {
PARTS:
DMux8Way(in=load, sel=address[6..8],
a=l0, b=l1, c=l2, d=l3, e=l4, f=l5, g=l6, h=l7);
RAM64(in=in, load=l0, address=address[0..5], out=o0);
RAM64(in=in, load=l1, address=address[0..5], out=o1);
RAM64(in=in, load=l2, address=address[0..5], out=o2);
RAM64(in=in, load=l3, address=address[0..5], out=o3);
RAM64(in=in, load=l4, address=address[0..5], out=o4);
RAM64(in=in, load=l5, address=address[0..5], out=o5);
RAM64(in=in, load=l6, address=address[0..5], out=o6);
RAM64(in=in, load=l7, address=address[0..5], out=o7);
Mux8Way16(a=o0, b=o1, c=o2, d=o3, e=o4, f=o5, g=o6, h=o7,
sel=address[6..8], out=out);
}
Chip RAM512
Eval
Reset
Clock: 81
Input pins
in
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
load
0
address
1 1 1 1 0 1 0 1 0
dec
Output pins
out
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
Internal pins
l0
0
l1
0
l2
0
Test
Steps: 292 Outputs: 320
Slow
Fast
Test Script
Compare File
Output File
Diff Table
set address %B101101010,
eval,
output;
set address %B110101010,
eval,
output;
set address %B111101010,
eval,
output;
Simulation successful: The output file is identical to the compare file
ENG
IN
23:01
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HDL
Builtin
Project 3
RAM4K
CHIP RAM4K {
PARTS:
DMux8Way(in=load, sel=address[9..11],
a=l0, b=l1, c=l2, d=l3, e=l4, f=l5, g=l6, h=l7);
RAM512(in=in, load=l0, address=address[0..8], out=o0);
RAM512(in=in, load=l1, address=address[0..8], out=o1);
RAM512(in=in, load=l2, address=address[0..8], out=o2);
RAM512(in=in, load=l3, address=address[0..8], out=o3);
RAM512(in=in, load=l4, address=address[0..8], out=o4);
RAM512(in=in, load=l5, address=address[0..8], out=o5);
RAM512(in=in, load=l6, address=address[0..8], out=o6);
RAM512(in=in, load=l7, address=address[0..8], out=o7);
Mux8Way16(a=o0, b=o1, c=o2, d=o3, e=o4, f=o5, g=o6, h=o7,
sel=address[9..11], out=out);
}
Chip RAM4K
Eval
Reset
Clock: 81
Input pins
in
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
load
0
address
1 1 1 1 0 1 0 1 0 1 0 1
dec
Output pins
out
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
Internal pins
l0
0
l1
0
l2
0
Test
Steps: 292 Outputs: 320
Slow
Fast
Test Script
Compare File
Output File
Diff Table
output;
set address %B101101010101,
eval,
output;
set address %B110101010101,
eval,
output;
set address %B111101010101,
eval,
output;
Simulation successful: The output file is identical to the compare file
ENG
IN
23:02
22-10-2026


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HDL
Builtin
Project 3
RAM8
1 CHIP RAM8 {
2 IN in[16], load, address[13];
3 OUT out[16];
4
5 PARTS:
6 DMux(in=load, sel=address[12], a=la, b=lb);
7
8 RAM4K(in=in, load=la, address=address[0..11], out=oa);
9 RAM4K(in=in, load=lb, address=address[0..11], out=ob);
10
11 Mux16(a=oa, b=ob, sel=address[12], out=out);
12 }
Chip RAM8
Eval
Reset
Clock: 46
Input pins
in
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
load
0
address
0 0 0 0 0 0 0 0 0 0 1 1 1
dec
Output pins
out
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
Internal pins
la
0
lb
0
oa
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
Test
Steps: 160 Outputs: 173
Slow
Fast
Test Script
Compare File
Output File
Diff Table
550 set address 5,
551 eval,
552 output;
553 set address 6,
554 eval,
555 output;
556 set address 7,
557 eval,
558 output;
559
Simulation successful: The output file is identical to the compare file
ENG
IN
23:03
22-10-2026


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HDL
Builtin
Project 3
RAM16K
CHIP RAM16K {
IN in[16], load, address[14];
OUT out[16];
PARTS:
DMux4Way(in=load, sel=address[12..13], a=l0, b=l1, c=l2, d=l3);
RAM4K(in=in, load=l0, address=address[0..11], out=o0);
RAM4K(in=in, load=l1, address=address[0..11], out=o1);
RAM4K(in=in, load=l2, address=address[0..11], out=o2);
RAM4K(in=in, load=l3, address=address[0..11], out=o3);
Mux4Way16(a=o0, b=o1, c=o2, d=o3, sel=address[12..13], out=out);
}
Chip RAM16K
Eval
Reset
Clock: 81
Input pins
in
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
load
0
address
1 1 1 1 0 1 0 1 0 1 0 1 0 1
dec
Output pins
out
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
Internal pins
l0
0
l1
0
l2
0
Test
Steps: 292 Outputs: 320
Slow
Fast
Test Script
Compare File
Output File
Diff Table
output;
set address %B10110101010101,
eval,
output;
set address %B11010101010101,
eval,
output;
set address %B11110101010101,
eval,
output;
ollama app.exe
Update available
Ollama version v0.40.8 is ready to install
Simulation successful: The output file is identical to the compare file
ENG
IN
23:08
22-10-2026